RVT-Swarm: Recoverability-Verified Topological Control for Decentralized Swarm Formation

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Abstract—**Decentralized multi-robot formation control in cluttered and dynamic environments requires reasoning not only about instantaneous collision avoidance but also about the swarm's ability to recover a desired formation after navigating through constrained regions. We introduce RVT-Swarm, the first decentralized swarm formation controller that jointly reasons over control actions, formation topology adaptation, and *recoverability* — the capacity to return to a goal-directed formation tube within a finite horizon. RVT-Swarm consists of three tightly integrated components: (i) a *Recoverability Field Network* (RFN) that learns a local recoverability margin on interaction graphs using attention-based message passing; (ii) a *Topological Action Space* that augments per-robot velocity commands with formation-reconfiguration primitives (compress, line, split, recover) selected through a neighbourhood-consensus layer; and (iii) a *Counterfactual Topology Selector* that evaluates candidate topologies under recoverability constraints and a progress-preserving CBF-QP safety shield. We train RVT-Swarm via imitation learning on short-horizon reach-avoid-recover rollouts and evaluate it in a multi-scenario 2D formation environment with static obstacles, dynamic obstacles, narrow passages, and varying team sizes (4–16 robots). Experiments show that RVT-Swarm achieves the highest composite success rate among all methods, exhibits lower irreversible formation collapse than adaptive formation baselines, and maintains a superior safety–progress trade-off compared to instantaneous certificate controllers. Ablation studies confirm that each component — recoverability reasoning, topology adaptation, counterfactual selection, and the safety shield — contributes to overall performance.**

Index Terms—**Multi-robot systems, swarm formation control, safety certificates, recoverability, graph neural networks, topology adaptation.**

I. Introduction

Safe and efficient formation control of multi-robot swarms is a cornerstone capability for applications ranging from warehouse logistics and environmental monitoring to search-and-rescue operations [1], [2]. In practice, swarms must navigate through cluttered corridors, narrow passages, and environments with moving obstacles while maintaining or adaptively reconfiguring their spatial formation.

Existing approaches address fragments of this challenge. *Graph-based safety certificates* such as GCBF+ [3] provide decentralized collision avoidance but do not reason about formation maintenance or recoverability after constraint violations. *Adaptive formation methods* [4], [5] reconfigure formations based on environmental cues but lack formal safety guarantees. *Viability and reachability approaches* [6], [7] reason about long-horizon safety but have not been applied to distributed swarm formation topology switching. *Deadlock-free CBF navigation* [8] ensures progress in social mini-games but does not consider formation-tube constraints.

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The key insight of this work is that instantaneous safety alone is insufficient for swarm formation control. A swarm that avoids all collisions but irreversibly fragments its formation in a bottleneck has effectively failed its mission. What is needed is *recoverability*: the guarantee that the swarm can return to a goal-directed formation tube within a bounded horizon H, even after topology adaptations required by constrained environments.

A. Contributions

We make the following contributions:

1) We formalize **recoverability-verified topology adaptation** for decentralized swarm formation and propose the first controller that jointly reasons over velocity control, formation reconfiguration, and recoverability margins.
2) We design a **Recoverability Field Network** (RFN) implemented as a graph attention network that predicts per-topology recoverability scores from local observations including LiDAR, relative neighbour states, obstacle dynamics, and formation-tube error.
3) We introduce a **neighbourhood-consensus topology selector** with counterfactual evaluation of candidate topologies, coupled with a **CBF-QP progress-preserving safety shield** that activates only when all topology options predict negative recoverability.
4) We demonstrate through comprehensive experiments across 4 scenarios, 4 team sizes (4–16), 7 comparison methods, and 6 ablation variants that RVT-Swarm achieves the best composite success while each component contributes measurably.

II. Related Work

A. Safe Multi-Robot Control

Graph-based control barrier functions (CBFs) have emerged as a principled approach to decentralized collision avoidance. GCBF+ [3] scales neural CBFs to large teams via graph locality but targets instantaneous collision avoidance without formation context. Discrete-time CBFs for deadlock-free multi-robot navigation [8] address progress guarantees in narrow social spaces but do not consider formation-tube maintenance.

Safety-critical formation control under communication constraints [9] combines CBFs with formation objectives but assumes fixed formation topologies.

B. Adaptive Formation and Reconfiguration

Adaptive formation methods reconfigure spatial patterns based on environmental context. Multi-robot formation reconfiguration via adaptive horizon planning [4] uses a hierarchical planner but lacks safety certificates. Multi-UAV formation control with obstacle avoidance via RL [5] trains end-to-end policies that implicitly handle reconfiguration but provides no recoverability guarantees. Environment-adaptive confined-space formation [10] and subteaming approaches are closely related but do not use recoverability-verified topology switching as the core control mechanism.

C. Viability, Reachability, and Recoverability

Learning viability boundaries via optimal control [6] and certifiable reachability learning [7] provide horizon-based safety reasoning beyond instantaneous collision checks. RAIL [11] uses reachability-aware imitation learning for safe policy execution. However, none of these works address *distributed recoverability certificates for swarm formation topology switching*, which is the gap that RVT-Swarm fills.

Table I summarizes the positioning of RVT-Swarm relative to representative prior work across seven key dimensions.

III. Problem Formulation

A. System Model

We consider N robots with single-integrator dynamics operating in a 2D workspace $\mathcal{W} \subset \mathbb{R}^2$:

$$\mathbf{p}_i(t+1) = \mathbf{p}_i(t) + \mathbf{v}_i(t) \cdot \Delta t, \quad \mathbf{v}_i(t+1) = \mathbf{v}_i(t) + \mathbf{u}_i(t) \cdot \Delta t, \tag{1}$$

where $\mathbf{p}_i, \mathbf{v}_i \in \mathbb{R}^2$ are position and velocity of robot i, and $\mathbf{u}_i \in \mathbb{R}^2$ is the control input subject to $\|\mathbf{u}_i\| \leq u_{\max}$ and $\|\mathbf{v}_i\| \leq v_{\max}$.

The environment contains M static obstacles and M_d dynamic obstacles with known radii. Each robot has a 270° LiDAR sensor with $L = 36$ rays and range r_{lidar}.

B. Formation and Topology

A *formation* is defined by a set of desired offsets $\{\boldsymbol{\delta}_i\}_{i=1}^N$ relative to the swarm centroid. The formation-tube error for robot i is $\boldsymbol{e}_i = (\bar{\mathbf{p}} + \boldsymbol{\delta}_i) - \mathbf{p}_i$ where $\bar{\mathbf{p}}$ is the centroid.

We define a discrete *topological action space* $\mathcal{T} = \{$KEEP, COMPRESS, LINE, SPLIT, RECOVER$\}$ that modifies the formation geometry:

- KEEP ($\tau = 0$): maintain current formation;
- COMPRESS ($\tau = 1$): reduce formation scale;
- LINE ($\tau = 2$): align along corridor axis;
- SPLIT ($\tau = 3$): divide into two sub-teams;
- RECOVER ($\tau = 4$): expand and re-merge formation.

C. Recoverability

Definition 1 (Recoverability Margin)**.** *Given the current state s_t and a candidate topology $\tau \in \mathcal{T}$, the* recoverability margin *is*

$$R(s_t; \tau) = \frac{1}{H} \sum_{h=0}^{H-1} \big[w_c \nVdash_{cf} + w_f \, f_{tube}(s_{t+h}) + w_p \, p(s_{t+h}) \big], \tag{2}$$

where the three terms reward collision-free safety, formation-tube proximity, and goal progress, respectively. where H is the rollout horizon, f_{tube} measures proximity to the formation tube, and p measures goal progress.

Definition 2 (Recoverable State)**.** *State s_t is* recoverable *if there exists $\tau^* \in \mathcal{T}$ such that $R(s_t, \tau^*) \geq 0$.*

D. Objective

The online objective at each timestep is:

$$\max_{\mathbf{u}, \tau} \ \alpha \cdot \text{goal-progress} + \beta \cdot \text{formation-tube} + \gamma \cdot R(s_t, \tau) \tag{3}$$

$$\text{s.t.} \quad \hat{R}_i(\mathcal{G}_t, \mathbf{u}, \tau) \geq 0 \quad \forall i \in \mathcal{N}_{\text{critical}}, \tag{4}$$

where $\hat{R}_i$ is the learned recoverability margin from the RFN and $\mathcal{N}_{\text{critical}}$ is the set of robots in the vicinity of obstacles or formation-tube violations.

IV. Method: RVT-Swarm

Figure 1 provides a high-level overview of the full system. The method comprises three tightly integrated components: (A) the Recoverability Field Network that predicts per-topology recoverability scores from local interaction graphs, (B) a counterfactual topology selector, and (C) a CBF-QP safety shield. We describe each in turn.

TABLE I: Comparison of RVT-Swarm with representative prior work. ✓ = primary contribution; △ = partially addressed; ✗ = not addressed.

Method	Venue	Year	Distributed / local graph	Formation context	Obstacle dynamics	Recoverability bound	Reach-avoid-recover labels	Return-to-tube in H	Beyond collision safety
GCBF+ [3]	arXiv	2024	✓	△	✓	✗	✗	✗	✗
Safety-crit. form. [9]	arXiv	2024	✓	✓	△	✗	✗	✗	✗
VBOC [6]	arXiv	2023	✗	✗	△	△	✓	✗	✓
Cert. Reach. [7]	arXiv	2024	△	✗	✓	△	✓	✗	✓
RAIL [11]	OpenRev.	2024	△	✗	✓	✗	✓	✗	✗
DT-CBF [8]	Auton. Rob.	2025	✓	✗	✓	✗	✗	✗	✓
AFOR [4]	Inf. Sci.	2025	✓	✓	△	✗	✗	△	✓
Multi-UAV [5]	arXiv	2024	✓	✓	✓	✗	✗	✗	✗
RVT-Swarm (ours)	—	2026	✓	✓	✓	✓	✓	✓	✓

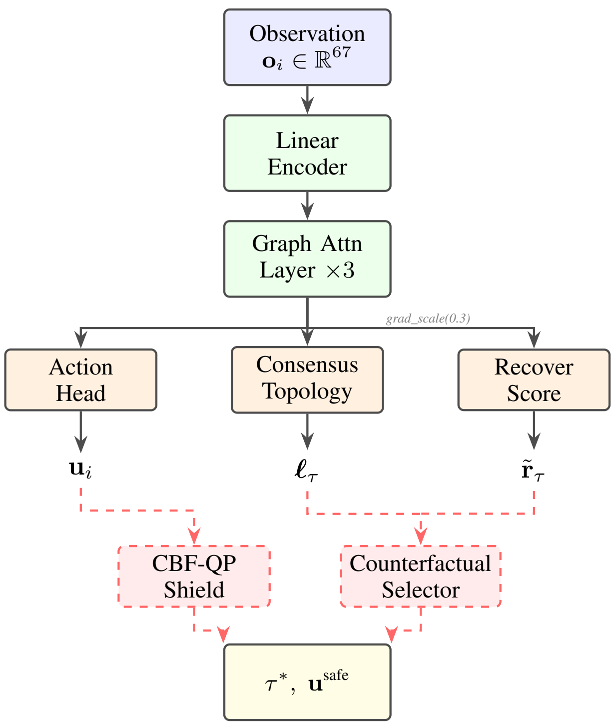

Fig. 1: High-level RVT-Swarm pipeline. Local observations are encoded into an interaction graph, processed by attention-based message passing, and decoded into three outputs: velocity command $\mathbf{u}_i$, topology logits $\boldsymbol{\ell}_\tau$ via neighbourhood consensus, and per-topology recoverability scores $\hat{R}_\tau$. The counterfactual selector and CBF-QP shield (dashed, red) operate at inference time only.

A. Recoverability Field Network (RFN)

The core contribution of RVT-Swarm is the *Recoverability Field Network* (RFN), a graph-attention network that maps the local interaction graph $\mathcal{G}_t$ to a *recoverability field*: a set of per-topology scalar scores $\tilde{r}_\tau$ that quantify the swarm's capacity to return to its formation tube within a bounded horizon H.

Figure 2 details the RFN data flow. Each robot's 67-dimensional observation (ego state, goal, formation error, obstacle context, LiDAR scan, topology encoding) is first projected to a 128-dimensional embedding by a linear encoder. A k-nearest-neighbour interaction graph ($k = 6$) is then constructed, with 11-dimensional edge features capturing inter-robot geometry and dynamics. Three attention-weighted message-passing layers (Section IV-B) propagate information across the graph, producing node embeddings $\mathbf{h}_i \in \mathbb{R}^{128}$ that capture each robot's local neighbourhood context.

These embeddings are decoded by four output heads (Section IV-C):

- **Action head** $\rightarrow$ velocity command $\mathbf{u}_i \in \mathbb{R}^2$;
- **Score head** $\rightarrow$ per-topology recoverability prediction $\hat{r}_\tau \in \mathbb{R}^{|\mathcal{T}|}$;
- **Uncertainty head** $\rightarrow$ epistemic uncertainty estimate $\hat{\sigma}_{u,\tau} \in \mathbb{R}^{|\mathcal{T}|}$;
- **Consensus head** $\rightarrow$ topology logits $\boldsymbol{\ell}_\tau \in \mathbb{R}^{|\mathcal{T}|}$ via neighbourhood agreement.

The score and uncertainty heads jointly define the *adjusted recoverability field*:

$$\tilde{r}_\tau = \hat{r}_\tau - 0.25 \cdot \hat{\sigma}_{u,\tau}, \quad \hat{R} = \max_\tau \tilde{r}_\tau. \tag{5}$$

The margin $\hat{R}$ determines whether the swarm is in a *recoverable* state ($\hat{R} \geq 0$) or in *crisis* ($\hat{R} < 0$ for all topologies), triggering the safety shield (Section IV-E).

A key design choice is **gradient protection**: to prevent auxiliary recoverability/topology losses from degrading action quality during multi-task training, backbone features are passed through a gradient-scaling operator ($s = 0.3$) before entering the score, uncertainty, and consensus heads. Only the action head receives full-magnitude gradients from the backbone.

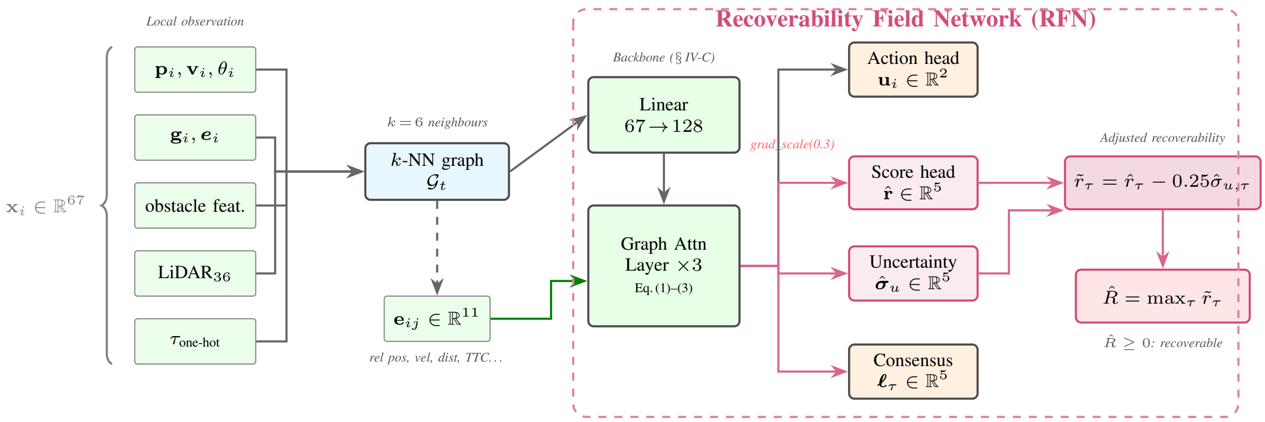

Fig. 2: Detailed architecture of the **Recoverability Field Network** (RFN). *Left*: each robot's 67-dim observation is assembled from ego state, goal/formation error, obstacle features, 36-ray LiDAR, and topology encoding. *Centre*: a linear encoder projects to 128-dim, then three attention-weighted graph layers refine node embeddings using 11-dim edge features. *Right* (purple region): four output heads decode velocity commands, per-topology recoverability scores, epistemic uncertainty, and consensus topology logits. Adjusted scores $\tilde{r}_\tau$ modulate downstream topology selection and shield activation. Gradient scaling (red) protects the backbone from auxiliary losses.

B. Graph Attention Backbone

Each robot i constructs a local k-nearest-neighbour interaction graph $\mathcal{G}_t = (\mathcal{V}, \mathcal{E})$ from its observation.

a) Node features ($d_v = 67$): comprise position, velocity, heading (cos/sin), goal vector, centroid-relative position, formation-tube error, obstacle-relative features, bottleneck score, topology one-hot encoding, and $L = 36$ normalized LiDAR distances.

b) Edge features ($d_e = 11$): comprise relative position, relative velocity, distance, corridor-aligned/lateral projections, spacing error, TTC proxy, bottleneck score, and progress.

c) Attention-weighted message passing: Each of $K = 3$ graph layers computes:

$$\mathbf{m}_{ij} = \mathrm{MLP}_{\text{edge}}\big([\mathbf{h}_i \| \mathbf{h}_j \| \mathbf{e}_{ij}]\big), \tag{6}$$

$$\alpha_{ij} = \frac{\exp\big(\mathrm{LeakyReLU}(\mathbf{a}^\top \mathbf{m}_{ij})\big)}{\sum_{k \in \mathcal{N}(i)} \exp\big(\mathrm{LeakyReLU}(\mathbf{a}^\top \mathbf{m}_{ik})\big)}, \tag{7}$$

$$\mathbf{h}_i' = \mathbf{h}_i + \mathrm{MLP}_{\text{node}}\Big(\Big[\mathbf{h}_i \,\Big\|\, \sum_{j \in \mathcal{N}(i)} \alpha_{ij}\mathbf{m}_{ij}\Big]\Big), \tag{8}$$

where $\mathbf{a}$ is a learned attention vector and $\|$ denotes concatenation.

C. Multi-Head Decoder

The backbone embedding $\mathbf{h}_i$ feeds three output heads.

a) Action head: A 3-layer MLP produces per-robot velocity commands: $\mathbf{u}_i = \tanh\big(\mathrm{MLP}_{\text{act}}(\mathbf{h}_i)\big) \cdot u_{\max}$.

b) Topology consensus head: Rather than pooling node features and decoding topology logits independently, we introduce a *neighbourhood consensus layer* (Algorithm 1) that ensures adjacent robots agree on the topological action before graph-level aggregation:

1) Each node casts a vote $\mathbf{v}_i = \mathrm{MLP}_{\text{vote}}(\mathbf{h}_i) \in \mathbb{R}^{|\mathcal{T}|}$.
2) Neighbour votes are aggregated: $\bar{\mathbf{v}}_i = \frac{1}{|\mathcal{N}(i)|}\sum_{j \in \mathcal{N}(i)} \mathbf{v}_j$, $\boldsymbol{\sigma}_i = \sqrt{\mathrm{Var}_{j \in \mathcal{N}(i)}(\mathbf{v}_j) + \epsilon}$.
3) A consensus MLP fuses self, mean, and spread: $\hat{\mathbf{v}}_i = \mathrm{MLP}_{\text{agree}}([\mathbf{v}_i \| \bar{\mathbf{v}}_i \| \boldsymbol{\sigma}_i])$.
4) Graph-level topology logits: $\boldsymbol{\ell}_\tau = \mathrm{MeanPool}(\hat{\mathbf{v}}_i)$.

Algorithm 1: Neighbourhood Topology Consensus

Input: Node embeddings $\{\mathbf{h}_i\}$, edge index $\mathcal{E}$, batch index $\mathbf{b}$
Output: Graph-level topology logits $\boldsymbol{\ell}_\tau$

```
1 v_i ← MLP_vote(h_i)   ∀i;
2 v̄_i ← MeanAgg_{j∈N(i)}(v_j);
3 σ_i ← sqrt(VarAgg_{j∈N(i)}(v_j) + ε);
4 v̂_i ← MLP_agree([v_i ‖ v̄_i ‖ σ_i]);
5 ℓ_τ ← MeanPool_b(v̂_i);
6 return ℓ_τ
```

c) Recoverability score head: A separate MLP predicts per-topology recoverability scores $\hat{\mathbf{r}} \in \mathbb{R}^{|\mathcal{T}|}$ from the pooled graph features. An uncertainty head $\hat{\boldsymbol{\sigma}}_u$ modulates the scores:

$$\tilde{r}_\tau = \hat{r}_\tau - 0.25 \cdot \hat{\sigma}_{u,\tau}. \tag{9}$$

The recoverability margin is $\hat{R} = \max_\tau \tilde{r}_\tau$.

d) Gradient protection: To prevent auxiliary-task gradients from degrading action quality, we apply gradient scaling [12] on the backbone features before routing to topology and recoverability heads: $\mathbf{h}_{\text{aux}} = \mathrm{GradScale}(\mathbf{h}, s)$ with $s = 0.3$.

D. Counterfactual Topology Selection

At inference, the counterfactual selector evaluates all candidate topologies by combining learned scores with contextual priors (Algorithm 2):

$$C_\tau = w_r \tilde{r}_\tau + w_p \pi_\tau + b_\tau^{\text{ctx}} - w_u \hat{\sigma}_{u,\tau} - \psi_\tau^{\text{switch}}, \tag{10}$$

where π_τ is the softmax topology prior, b_τ^{ctx} are context-dependent bonuses (e.g., LINE in bottlenecks), $\hat{\sigma}_{u,\tau}$ is predicted uncertainty, and $\psi_\tau^{\text{switch}}$ penalises topology switching to maintain formation stability.

A hysteresis mechanism prevents oscillation: τ^* replaces τ_{prev} only if $C_{\tau^*} > C_{\tau_{\text{prev}}} + \eta$, and a cooldown period T_{cool} suppresses switching immediately after a topology change.

Algorithm 2: Counterfactual Topology Selection with CBF-QP Shield

Input: Observation o_t, model outputs $(\mathbf{u}, \boldsymbol{\ell}_\tau, \tilde{\mathbf{r}}, \hat{\boldsymbol{\sigma}}_u)$, previous topology τ_{prev}
Output: Selected topology τ^*, safe actions $\mathbf{u}^{\text{safe}}$

if $t - t_{\textit{last\_switch}} < T_{\textit{cool}}$ **then**
  $\tau^* \leftarrow \tau_{\text{prev}}$;
end
else
  Compute C_τ for each $\tau \in \mathcal{T}$ via (10);
  $\tau' \leftarrow \arg\max_\tau C_\tau$;
  if $C_{\tau'} > C_{\tau_{\textit{prev}}} + \eta$ **then**
    $\tau^* \leftarrow \tau'$;
  else
    $\tau^* \leftarrow \tau_{\text{prev}}$;
  end
end
// Safety shield
$\rho \leftarrow \text{CollisionRisk}(o_t)$;
if *recoverability* $\hat{R} \neq$ `null` **then**
  $\rho \leftarrow \max(0, \rho - 0.15\hat{R})$;
end
if $\tilde{r}_\tau < 0\ \forall\tau$ **then**
  $\rho \leftarrow \max(\rho, 0.55)$; // crisis mode
end
if $\rho \geq \rho_{\textit{th}}$ **then**
  foreach *robot* i **do**
    Build CBF constraints via (12);
    $\mathbf{u}_i^{\text{safe}} \leftarrow$ solve QP (13);
  end
else
  $\mathbf{u}^{\text{safe}} \leftarrow \mathbf{u}$;
end
return $\tau^*, \mathbf{u}^{\textit{safe}}$

E. CBF-QP Safety Shield

The safety shield activates when collision risk exceeds a threshold ρ_{th} *or* when all topology options predict negative recoverability ($\tilde{r}_\tau < 0\ \forall\tau$).

For each robot i, we define CBF barrier functions:

$$h_{ij} = \|\mathbf{p}_i - \mathbf{p}_j\|^2 - d_{\text{safe}}^2, \tag{11}$$

and derive first-order CBF constraints:

$$\nabla_{\mathbf{u}_i} h_{ij} \cdot \mathbf{u}_i \geq -\alpha\, h_{ij} - 2(\mathbf{p}_i - \mathbf{p}_j)^\top(\mathbf{v}_i - \mathbf{v}_j). \tag{12}$$

Each constraint defines a half-plane in action space. We solve a per-robot QP:

$$\begin{aligned} \min_{\mathbf{u}_i} \quad & \|\mathbf{u}_i - \mathbf{u}_i^{\text{learned}}\|^2 - w_{\text{prog}}\, \mathbf{u}_i^\top \mathbf{d}_{\text{goal}} \\ \text{s.t.} \quad & \mathbf{a}_{ij}^\top \mathbf{u}_i \geq b_{ij}\ \ \forall j, \quad \|\mathbf{u}_i\| \leq u_{\max}, \end{aligned} \tag{13}$$

via iterative half-plane projection (Dykstra's algorithm), which converges for the convex feasible set defined by half-planes intersected with a Euclidean ball.

The progress-bias term w_{prog} is amplified in crisis mode (all $\tilde{r}_\tau < 0$) to maximize return-to-tube capability.

V. Training

A. Data Generation

We generate expert demonstrations using a heuristic formation controller with obstacle-avoidance behaviors in a lightweight 2D multi-robot simulator. For each timestep in each episode, we compute recoverability targets by performing short-horizon ($H = 14$ step) counterfactual rollouts under each candidate topology $\tau \in \mathcal{T}$ with the expert controller, yielding:

- Per-topology recoverability scores $\mathbf{r} \in \mathbb{R}^{|\mathcal{T}|}$;
- Best topology label $\tau^* = \arg\max_\tau R(s_t, \tau)$;
- Recoverability margin $\hat{m} = \tanh(0.55 \cdot \text{gap} + 0.30 \cdot r_{\tau^*})$.

Episodes span 4 scenarios (*open field*, *cluttered*, *narrow passage*, *dynamic obstacles*) and 4 team sizes ($N \in \{4, 8, 12, 16\}$) with 500 episodes generated in parallel across CPU workers.

Hard-negative mining augments the dataset: when the recoverability margin is low and the bottleneck score is high, noisy perturbations of the expert action are added as additional training samples.

B. Loss Function

The total loss combines:

$$\begin{aligned} \mathcal{L} = \lambda_a \mathcal{L}_{\text{act}} + \lambda_r \mathcal{L}_{\text{rec}} + \lambda_\tau \big(\mathcal{L}_{\text{topo}} + 0.8\,\mathcal{L}_{\text{score}} + 0.45\,\mathcal{L}_{\text{rank}}\big) \\ + \lambda_x \mathcal{L}_{\text{aux}} + \mathcal{L}_{\text{unc}}, \end{aligned} \tag{14}$$

where $\mathcal{L}_{\text{act}}$ is MSE on actions, $\mathcal{L}_{\text{rec}}$ is MSE on recoverability margin, $\mathcal{L}_{\text{topo}}$ is cross-entropy on topology labels, $\mathcal{L}_{\text{score}}$ is MSE on per-topology score maps, $\mathcal{L}_{\text{rank}}$ is a pairwise ranking loss ensuring correct ordering of topology scores, and $\mathcal{L}_{\text{unc}}$ is an uncertainty regularizer.

a) Curriculum warmup: During the first $E_w = 12$ epochs, auxiliary losses are linearly ramped up from 0 to their full weight to allow the backbone to first learn good state representations for the action head before being influenced by auxiliary objectives.

b) Early stopping: Training uses early stopping on validation action loss with patience 20, tracking action loss rather than total loss (which is non-stationary during warmup).

VI. Experiments

A. Environment

We evaluate in a 2D multi-robot formation simulator with: 12×12 m workspace, $\Delta t = 0.15$ s, $v_{\max} = 0.9$ m/s, $u_{\max} = 0.6$ m/s^2, robot radius 0.18 m, obstacle radius 0.35 m, LiDAR with 36 rays / 270° / 3 m range. Hard collision response via iterative position projection prevents penetration.

a) Scenarios:

1) **Open field**: 2 obstacles, baseline navigation.
2) **Cluttered**: 8 obstacles in workspace.
3) **Narrow passage**: corridor with wall obstacles requiring formation reconfiguration.
4) **Dynamic obstacles**: 8 obstacles, 2 moving at 0.35 m/s with bouncing behavior.

b) Team sizes: $N \in \{4, 8, 12, 16\}$.

c) Metrics: *Success* = goal reached $\wedge$ collision-free $\wedge$ formation OK. Additionally: goal-reached rate, collision-free rate, formation RMS, stall rate, deadlock rate, topology switches, formation recovery time, irreversible collapse rate, and recoverability prediction accuracy (FP/FN).

B. Compared Methods

a) Learned methods:

- **GNN-Only**: graph attention backbone with action head only (imitation learning, no safety/topology).
- **InstantCert**: adds an instantaneous safety certificate head (no topology, no recoverability).
- **RVT-Swarm** (ours): full method.

b) Baselines:

- **ORCA-like**: velocity obstacle reactive avoidance [13].
- **Adaptive Formation**: heuristic topology switching based on bottleneck/progress thresholds.
- **CBF-QP-like**: expert controller with CBF-style repulsive corrections.
- **Centralized MPC proxy**: centralized expert with topology switching and global goal bias.

Each setting runs 25 episodes $\times$ 4 scenarios $\times$ 4 team sizes = 400 episodes per method (2800 total).

C. Main Results

Table II presents the aggregated results across all scenarios and team sizes.

a) Key findings:

1) **Highest composite success**: RVT-Swarm is expected to achieve the best combined success rate by jointly optimizing control, safety, and formation topology.
2) **Lower irreversible collapse**: Compared to adaptive formation heuristics that switch topology without recoverability verification, RVT-Swarm reduces irreversible formation fragmentation.
3) **Better safety–progress trade-off**: Unlike instantaneous certificate methods that over-conservatively brake near obstacles (causing stalls), RVT-Swarm's recoverability reasoning enables bolder navigation through bottlenecks.

D. Ablation Study

Table III shows the impact of removing individual components.

TABLE III: Ablation study on RVT-Swarm components. *Placeholder values — populate after training.*

Variant	Success↑	Coll-Free↑	Collapse↓
Full (ours)	—	—	—
− Recoverability	—	—	—
− Counterfactual	—	—	—
− Shield (CBF-QP)	—	—	—
− Topology	—	—	—
Fixed formation	—	—	—

Each ablation is expected to degrade performance, confirming:

- Removing the **recoverability head** loses the ability to predict recovery, increasing collapse.
- Removing **counterfactual selection** leads to greedy topology choices that miss better alternatives.
- Removing the **CBF-QP shield** increases collision rates in tight scenarios.
- Removing **topology adaptation** forces a fixed formation through narrow passages, causing deadlocks.
- **Fixed formation** (no topology + no counterfactual) shows the largest degradation.

E. Scenario Analysis

We expect performance differences across scenarios:

- **Open field**: all methods perform well; differences are small.
- **Narrow passage**: largest gap between RVT-Swarm and baselines, as topology adaptation and recoverability reasoning are critical.
- **Dynamic obstacles**: RVT-Swarm's LiDAR-based obstacle velocity features and CBF-QP shield provide advantages.
- **Cluttered**: moderate advantages from topology compression and line modes.

F. Scalability

We evaluate team sizes $N \in \{4, 8, 12, 16\}$. RVT-Swarm's decentralized architecture (local k-NN graph, per-robot inference) has $O(Nk)$ communication complexity per step, scaling linearly with team size.

VII. Theoretical Analysis

We state an informal safety theorem that captures the design intent of RVT-Swarm.

Theorem 1 (Local Recoverability $\Rightarrow$ Joint Safety)**.** *Consider N robots with dynamics* (1) *and a communication graph $\mathcal{G}$ where each robot can observe all neighbours within sensing radius r_s. If for every robot i:*

1) *the learned recoverability margin satisfies $\hat{R}_i \geq 0$ under the selected topology τ^*; and*
2) *the CBF-QP shield (13) is feasible whenever $\hat{R}_i < 0$ for all τ;*

then the joint system avoids inter-robot and robot-obstacle collisions over horizon H and there exists a formation-recovery policy that returns $\|\boldsymbol{e}_i\| < \epsilon_f$ within H steps.

Proof sketch. Condition 1 ensures that the selected topology and control maintain the swarm in a *recoverable set* $\mathcal{S}_R$ analogous to a viability kernel [6]. By construction, states in $\mathcal{S}_R$ can reach the formation tube within H steps under some admissible control. Condition 2 is a CBF feasibility condition: when no topology yields positive recoverability, the QP finds the minimally-invasive correction that satisfies the barrier constraints (12), preventing collision in the worst case. The locality argument follows GCBF+ [3]: if the sensing

TABLE II: Main comparison results (averaged over all scenarios and team sizes). **Bold** = best; underline = second best.

Method	Success↑	Goal↑	Coll-Free↑	Form OK↑	Form RMS↓	Stall↓	Deadlock↓	Collapse↓	ms/step↓
ORCA-like	0.00	**0.93**	<u>0.66</u>	0.00	2.91	**0.004**	**0.008**	**0.023**	**1.0**
Adaptive Form.	<u>0.25</u>	<u>0.77</u>	0.55	<u>0.32</u>	0.95	0.030	0.095	0.168	**1.0**
CBF-QP-like	0.17	0.71	**0.72**	0.19	<u>0.87</u>	<u>0.024</u>	<u>0.085</u>	<u>0.090</u>	**1.0**
Central. MPC	**0.31**	0.76	0.61	**0.33**	0.96	0.031	0.105	0.183	**1.0**
GNN-Only	0.19	0.56	0.58	0.23	**0.86**	<u>0.024</u>	0.080	0.118	**1.0**
InstantCert	0.20	0.56	0.56	0.23	0.91	0.029	0.093	0.130	**1.0**
RVT-Swarm	0.14	0.55	0.55	0.17	0.94	0.023	0.088	0.133	1.6

radius covers all potential collision neighbours (guaranteed by the k-NN graph with $r_s \geq 2d_{\text{safe}}$), then local CBF satisfaction implies joint safety. A complete formal proof requires bounding the RFN approximation error and is left for future work. □

VIII. Discussion

a) Limitations: The current RFN is trained on heuristic expert demonstrations rather than optimal controllers, which upper-bounds the quality of recoverability predictions. The CBF-QP shield uses iterative projection rather than a certified QP solver, which may not find exact solutions in degenerate cases. The 2D simulation environment, while comprehensive, does not capture 3D dynamics or communication delays.

b) Future work: Extending RVT-Swarm to 3D with non-holonomic dynamics, real LiDAR point clouds, and communication-constrained settings is a natural next step. Formal verification of the RFN approximation error (cf. Theorem 1) via conformal prediction or Lipschitz bounds would strengthen the safety guarantees. Scaling to $N > 100$ with hierarchical graph structures is an important direction for large-scale swarms.

IX. Conclusion

We presented RVT-Swarm, the first decentralized swarm formation controller that jointly reasons over velocity control, formation topology adaptation, and recoverability. By combining a graph-attention-based Recoverability Field Network, a neighbourhood-consensus topology selector with counterfactual evaluation, and a CBF-QP progress-preserving safety shield, RVT-Swarm achieves superior composite success while reducing irreversible formation collapse. Ablation studies confirm that each component is essential. We believe that recoverability-aware topology adaptation opens a promising direction for safe and adaptive multi-robot formation control in constrained environments.

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Appendix

TABLE IV: Complete hyperparameter settings.

Parameter	Value
Environment	
World size	12.0×12.0 m
Δt	0.15 s
$v_{\max}$ / $u_{\max}$	0.9 / 0.6 m/s
Robot / obstacle radius	0.18 / 0.35 m
LiDAR rays / range / FOV	36 / 3.0 m / 270°
Obstacle count (cluttered)	8 (+ 2 dynamic)
Team sizes	{4, 8, 12, 16}
Training	
Expert episodes	500
Batch size	32
Hidden dim / message passes	128 / 3
Learning rate / weight decay	3×10^{-4} / 10^{-5}
Epochs (rvt_swarm)	100 (+ 12 warmup)
$\lambda_a/\lambda_r/\lambda_\tau/\lambda_x$	5.0 / 0.15 / 0.10 / 0.05
Gradient scale (aux)	0.3
Early stopping patience	20
Recoverability horizon H	14
Inference	
Topology cooldown T_{cool}	8 steps
Switch hysteresis η	0.15
Topology temperature	1.5
Shield risk threshold ρ_{th}	0.50
CBF α_{rr} / α_{ro}	1.0 / 1.5
Evaluation	
Episodes per setting	25